Exam Date & Time: 26-Dec-2024 (09:30 AM - 12:30 PM)

MANIPAL ACADEMY OF HIGHER EDUCATION

THIRD SEMESTER B.Tech. (ELECTRONICS & COMMUNICATION ENGINEERING) EXAMINATIONS - DECEMBER 2024 / JANUARY 2025
SUBJECT: ECE 2122/ECE_2122- NETWORK ANALYSIS

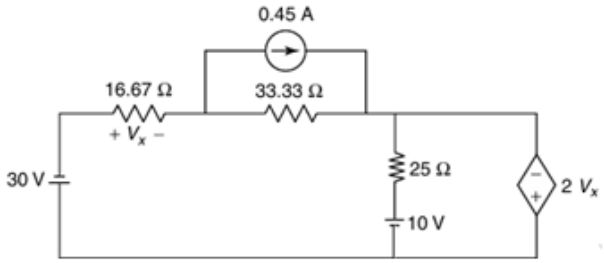
NETWORK ANALYSIS [ECE 2122]

Marks: 50 **Duration: 180 mins.**

A

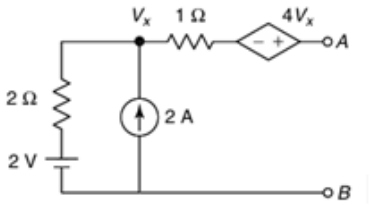
Answer all the questions.

1A) Find the voltage Vx in the network shown below figure.



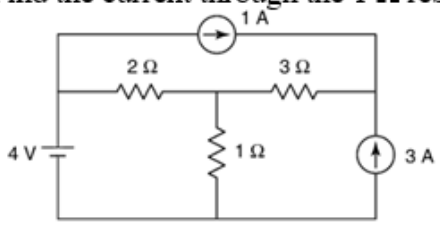
(4)

1B) Obtain the Thevenin equivalent network of figure shown below with respect to terminals A and B.



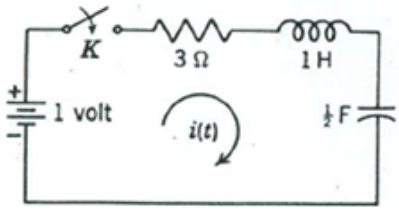
(4)

1C) Find the current through the 1 Ω resistor using superposition theorem in network shown below.



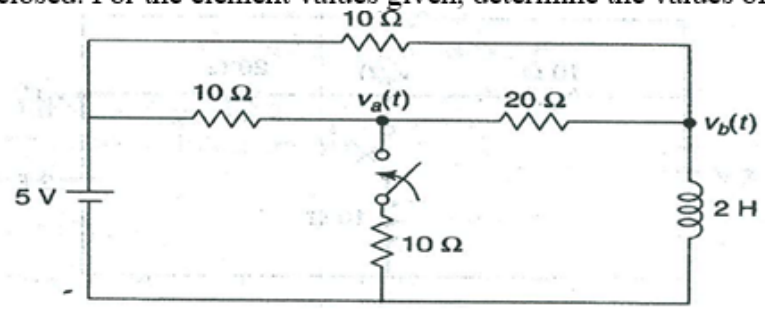
(2)

2A) In the network shown in figure, the switch K is closed at t = 0. For the numerical values given, find complementary, particular and total response i(t) using differential equation. Also draw the current response as a function of time.



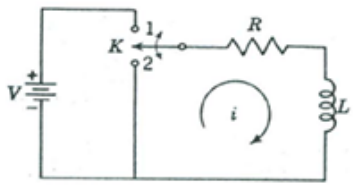
(5)

2B) In the network shown in Figure, a steady state is reached with the switch open. At t = 0, the switch is closed. For the element values given, determine the values of $v_a(0^-)$, $v_b(0^-)$, $v_a(0^+)$ and $v_b(0^+)$.



(3)

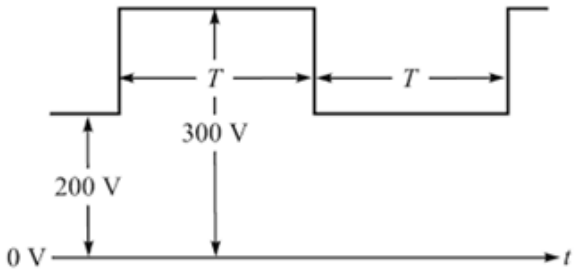
2C) In the network shown in Figure, the switch K is moved from position 1 to position 2 at t = 0, a steady state current having previously been established in the RL circuit. Find the general solution for the current *i*.



(2)

3A) A 10 Hz symmetrical square wave whose peak-to-peak amplitude is 2 V is fed to an amplifier. Calculate and sketch the output waveform if the lower 3-dB frequency is 0.3 Hz. (4)

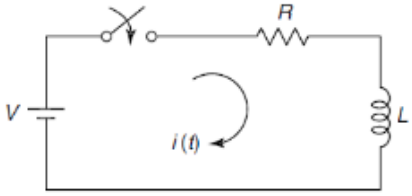
3B) The square wave shown in the figure is fed to an RC integrating circuit. Compute and plot the output waveform under steady-state conditions, if RC = 2.5T



(4)

3C) Prove that a High-pass RC Circuit is a Differentiator. (2)

4A) A general network consisting of a resistor and an inductor is shown in Fig. 4A. The switch is closed at time t = 0. Using Laplace transform, obtain the expression for i(t) for t > 0.



(4)

Fig: 4A

4B) (4)

In the network shown in Fig. 4B, the switch is opened at t = 0. Steady-state condition is achieved before t = 0. Using Laplace transform, calculate i(t).

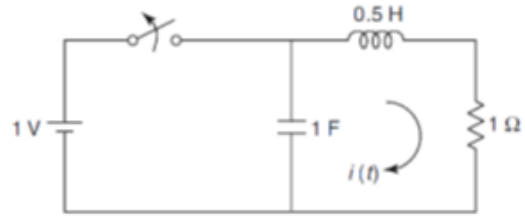
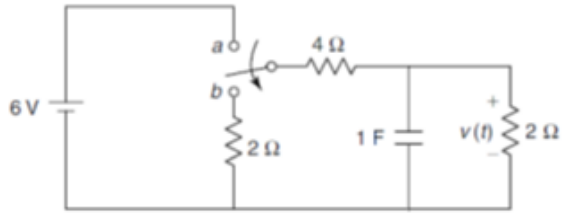


Fig: 4B

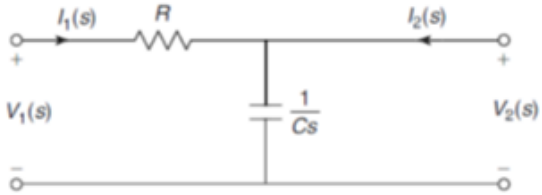
4C) In the network of Fig. 4C, the switch is moved from a to b, at t = 0. For t > 0, calculate v(t) using Laplace transform.



(2)

Fig: 4C

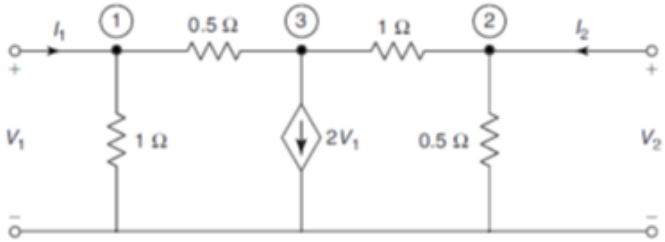
5A) Obtain the voltage transfer function of the two-port network shown in Fig. 5A



(3)

Fig: 5A

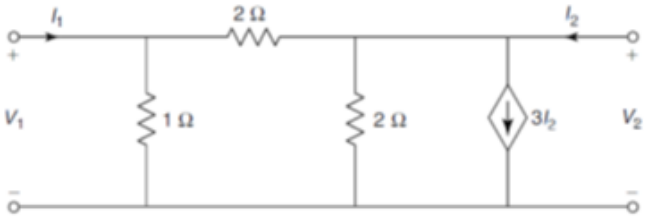
5B) Obtain the Y and Z-parameters of the network shown in Fig. 5B



(4)

Fig: 5B

5C) Obtain the ABCD parameters for the two-port network shown in Fig. 5C



(3)

Fig: 5C

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